

abstract way. With this approach, simple but powerful applications can have around 10 lines of high-level code.

Consensus of the past - Timestamping

A blockchain platform inherently maintains a consensus about the transactions or computations that have already occurred. Even though there is no central authority, it achieves this by distributing the authority among its users. Changes are publicly announced and incorporated by everyone in the order they are received. As the data is represented by the hash and all subsequent blocks, the blockchain also serves as a distributed timestamp server. Once the hash of a data block is incorporated into the blockchain, it will be replicated across the entire network. All successive blocks in the chain are dependent on this and since the hash is unique to the data, it is undeniable proof that the data existed at the time of incorporation into the blockchain. In essence, blockchain technology employs a linear cause and effect model to record events, and verifies truth by distributed consensus. What happened in the blockchain, stays on the blockchain!

Consensus of the future - Smart Contracts

However, this distributed consensus platform is not only useful for past events, but future events as well. In the blockchain we have a distributed ledger that cannot be changed. This is like a permanent record of your words. The words themselves, may be talking about the future, like making a promise! Promises and trust are very related, as trustworthy people are defined as those who keep their promises. Promises can also be viewed as future transactions, as the person making the promise gets something in return for keeping their promise. In the human world, we can make selfless promises, but we still obtain trust in return, if nothing else. With blockchain, the issue of accountability of promises is taken care of by the property of being distributed, but the issue of trust resides further away in time (when the promise is/isn't fulfilled) and cannot be immediately resolved by virtue of being distributed.

Enter the Turing complete blockchain. With the capability to build a program that lives in the blockchain, we can design 'Smart Contracts'. In the simplest case, imagine a transaction inside an if statement. Perhaps there is also a deadline in the condition, This transaction is only fulfilled when the conditions are met, and not otherwise, because that is the way a smart contract works. There is no need for a third party, or even for the two parties